MF 281: Probability Theory

Week 2: The axioms of probability

Three rules, and everything that follows from them

Dr Abdulzeid Yen Anafo

Department of Computing and Data Analytics
University of Mines and Technology, Tarkwa

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Lecture notes for students of MF 281 Probability Theory.

Week 2: The axioms of probability

Learning Objectives

By the end of this week you should be able to:

- state the three axioms, and say what each one rules out;
- derive the complement rule, monotonicity and the addition rule from them;
- decide whether a proposed assignment of numbers is a valid probability law;
- use the addition rule and inclusion–exclusion for two and three events;
- apply the union bound, and say when it is worth using;
- handle sample spaces that are infinite, countable or continuous, without pretending every outcome is equally likely;
- explain why probability zero does not mean impossible.

2.1 Why axioms, when counting worked

Last week every outcome was equally likely, and probability was the ratio $|A|/|\Omega|$. That carries you through dice and committees and no further. A drawing pin does not land point-up half the time. A machine does not fail on Tuesday with the same chance as on Monday. And when the sample space is infinite, $|A|/|\Omega|$ is not even arithmetic.

What we need is a short list of properties any sensible assignment must satisfy, general enough to cover all these cases, from which everything else can be proved. Kolmogorov wrote that list in 1933, and it is three lines long.

2.2 The probability space

Definition 2.1 (Probability space)**.** A triple $(\Omega, \mathcal{F}, P)$: the sample space Ω of outcomes, a collection $\mathcal{F}$ of subsets of Ω called events, and a function $P : \mathcal{F} \to \mathbb{R}$ assigning a probability to each event.

$\mathcal{F}$ must be closed under the operations we actually perform: it contains Ω, and if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$, and if $A_1, A_2, \dots \in \mathcal{F}$ then $\bigcup_i A_i \in \mathcal{F}$. Such a collection is called a σ-algebra. When Ω is finite or countable you may take $\mathcal{F}$ to be every subset and never think about it again. It matters only for continuous spaces, where, remarkably, not every subset of $[0, 1]$ can be assigned a sensible length.

The axioms

1. **Non-negativity.** $P(A) \geq 0$ for every event A.
2. **Normalisation.** $P(\Omega) = 1$.
3. **Countable additivity.** If $A_1, A_2, \dots$ are pairwise disjoint then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

That is all. Everything in the rest of this course is a consequence of those three lines, together with the definition of conditional probability in Week 4.

Notice what the axioms do *not* say. They do not say where the numbers come from: that is a question about the world, answered by symmetry, by data, or by judgement. They do not say outcomes are equally likely. And the third axiom is about *disjoint* events only, which is exactly why overlapping events need the addition rule below.

2.3 What follows immediately

Proposition 2.1 (Empty set). $P(\varnothing) = 0$.

Proof. Ω and $\varnothing, \varnothing, \ldots$ are pairwise disjoint with union Ω, so by the third axiom $P(\Omega) = P(\Omega) + \sum_{i=1}^{\infty} P(\varnothing)$. Since $P(\Omega) = 1$ is finite, the sum must be zero, and as every term is non-negative by the first axiom, each is zero. □

Proposition 2.2 (Finite additivity). *If $A_1, \ldots, A_n$ are pairwise disjoint then $P(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} P(A_i)$.*

Proof. Append $A_{n+1} = A_{n+2} = \cdots = \varnothing$ and apply countable additivity; the added terms contribute nothing by the previous result. □

Proposition 2.3 (Complement). $P(A^c) = 1 - P(A)$, *and consequently* $P(A) \leq 1$.

Proof. A and A^c are disjoint with union Ω, so $P(A) + P(A^c) = P(\Omega) = 1$. Since $P(A^c) \geq 0$, we get $P(A) \leq 1$. □

Proposition 2.4 (Monotonicity). *If $A \subseteq B$ then $P(A) \leq P(B)$.*

Proof. Write $B = A \cup (B \setminus A)$, a disjoint union, so $P(B) = P(A) + P(B \setminus A) \geq P(A)$. □

Theorem 2.5 (Addition rule). *For any two events,*

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Proof. Split the union into three disjoint pieces: $A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$. By finite additivity,

$$P(A \cup B) = P(A \setminus B) + P(A \cap B) + P(B \setminus A).$$

Also $A = (A \setminus B) \cup (A \cap B)$ disjointly, so $P(A \setminus B) = P(A) - P(A \cap B)$, and similarly for B. Substituting both gives the result. □

The subtraction is there because adding $P(A)$ and $P(B)$ counts the overlap twice. This is Week 1's inclusion and exclusion, now for probabilities rather than counts, and for the same reason.

Example 1: two faults

A machine has $P(A) = 0.5$ of a hydraulic fault during a shift and $P(B) = 0.4$ of an electrical fault, with $P(A \cap B) = 0.2$.

Solution

At least one fault:

$$P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7.$$

No fault at all: $1 - 0.7 = 0.3$. A hydraulic fault only: $P(A) - P(A \cap B) = 0.3$. Exactly one fault: $0.7 - 0.2 = 0.5$.

Note the check available at every step: the four pieces (neither, A only, B only, both) are $0.3 + 0.3 + 0.2 + 0.2 = 1.0$. If they had not summed to one, something was wrong.

2.3.1 Three events

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

Example 2: three faults

With $P(A) = 0.5$, $P(B) = 0.4$, $P(C) = 0.3$, $P(A \cap B) = 0.2$, $P(A \cap C) = 0.15$, $P(B \cap C) = 0.1$

and $P(A \cap B \cap C) = 0.05$:

$$P(A \cup B \cup C) = 0.5 + 0.4 + 0.3 - 0.2 - 0.15 - 0.1 + 0.05 = 0.80.$$

Solution

So one shift in five passes with no fault of any kind. The pattern of signs, alternating as the intersections grow, continues for any number of events, though in practice nobody uses it beyond three: by then the union bound below is usually enough.

2.3.2 The union bound

Dropping the subtractions can only increase the total, so for any events at all,

$$P\left(\bigcup_i A_i\right) \le \sum_i P(A_i).$$

Example 3: a bound that is good enough

A system has 20 components, each failing in a given month with probability 0.001. What is the probability that at least one fails?

Solution

Without knowing how the failures are related, the exact answer is unavailable. But

$$P(\text{at least one fails}) \le 20 \times 0.001 = 0.02.$$

Two percent, at worst. If that is acceptable, the analysis is finished, and no independence assumption was needed. This is why reliability engineers reach for the union bound before anything more sophisticated: it is crude, it is always true, and very often it settles the question.

2.4 Is this a valid probability law?

To check a proposed assignment, verify the axioms: every value non-negative, and the total over a partition of Ω equal to one.

Example 4: a law with a constant to find

A contractor submits $Y \in \{1, 2, 3, 4, 5, 6\}$ forms, and the probability of y forms is proportional to y, that is $p(y) = ky$.

Solution

Normalisation requires

$$\sum_{y=1}^{6} ky = k(1 + 2 + 3 + 4 + 5 + 6) = 21k = 1 \quad \Rightarrow \quad k = \tfrac{1}{21},$$

and every $p(y) = y/21 \ge 0$, so the law is valid. Then

$$P(Y \le 3) = \frac{1 + 2 + 3}{21} = \frac{6}{21} = \frac{2}{7}, \qquad P(2 \le Y \le 4) = \frac{2 + 3 + 4}{21} = \frac{9}{21} = \frac{3}{7}.$$

A second proposal

Suppose instead someone proposes $p(y) = y^2/50$. Then $\sum_{y=1}^{6} y^2 = 91$, so the total is $91/50 = 1.82 \neq 1$: not a probability law, and no amount of interpretation rescues it. The only valid constant would be $1/91$.

2.5 When the sample space is infinite

2.5.1 Countably infinite

Toss a coin until the first head. The outcome is the number of tosses, $\Omega = \{1, 2, 3, \dots\}$, which is infinite but countable. Assign $P(k) = (1/2)^k$.

Example 5: an infinite sample space that still sums to one

Check normalisation, then find the probability that the first head comes on an even toss.

Solution

A geometric series:

$$\sum_{k=1}^{\infty} \left(\tfrac{1}{2}\right)^k = \frac{1/2}{1 - 1/2} = 1.\checkmark$$

For an even first head, sum over $k = 2, 4, 6, \dots$, a geometric series with first term $1/4$ and ratio $1/4$:

$$P(\text{even}) = \frac{1/4}{1 - 1/4} = \frac{1}{3}.$$

Note what happened

Every individual outcome has positive probability, there are infinitely many of them, and the total is still one, because the terms shrink fast enough. Countable additivity is exactly the axiom that licenses adding infinitely many terms like this.

2.5.2 Continuous

Spin a pointer and record where it stops, as a fraction of a turn: $\Omega = [0, 1]$. Here no individual outcome can have positive probability, because uncountably many equal positive numbers cannot sum to one. Instead probability is assigned to intervals, by length:

$$P([a, b]) = b - a.$$

Example 6: probability zero is not impossibility

With the pointer above, $P([0.2, 0.5]) = 0.3$, while $P(\{0.5\}) = 0$ for the single point 0.5.

Solution

Yet the pointer does stop somewhere, and wherever it stops was an outcome of probability zero. So "probability zero" and "cannot happen" are different statements, and only in a finite sample space do they coincide.

This is also why $\mathcal{F}$ mattered: on $[0, 1]$ there exist subsets so badly behaved that no consistent length can be assigned to them, and the σ-algebra is how we quietly exclude them. You will not meet one in this course, or in any application.

2.6 Continuity of probability

Proposition 2.6. *If* $A_1 \subseteq A_2 \subseteq \cdots$ *then* $P(\bigcup_n A_n) = \lim_{n\to\infty} P(A_n)$, *and if* $A_1 \supseteq A_2 \supseteq \cdots$ *then* $P(\bigcap_n A_n) = \lim_{n\to\infty} P(A_n)$.

In words, probability behaves sensibly under limits: a shrinking sequence of events has a probability shrinking to that of the limiting event.

Example 7: shrinking intervals

With the pointer of Example 6, let $A_n = (0, 1/n]$. These are nested downwards, and $P(A_n) = 1/n \to 0$. Their intersection is empty, because no positive number is below every $1/n$, and

indeed $P(\varnothing) = 0$. The limit and the intersection agree, as the proposition promises.

2.7 Where the numbers come from

The axioms constrain probabilities; they do not supply them. Three honest sources:

- **Symmetry.** A fair die, a well-shuffled deck. Powerful, and limited to situations where the symmetry is real.
- **Frequency.** The proportion of past shifts with a fault. This needs the future to resemble the past, which is an assumption, not a theorem.
- **Judgement.** An engineer's considered degree of belief, which is the only option for a one-off event. The axioms apply unchanged, which is the subject of Bayesian inference.

Whichever source you use, state it. "$P = 0.001$" from a manufacturer's test of 10,000 units is a different claim from "$P = 0.001$" from an engineer's intuition, even though the arithmetic that follows is identical.

2.8 Summary

Week 2 in six lines

- Three axioms: non-negative, total one, additive over disjoint events.
- Everything else, complement, monotonicity, addition rule, is derived.
- Additivity needs disjointness; overlapping events need the subtraction.
- A valid law is non-negative and sums to one; check before computing.
- Infinite sample spaces are fine if the probabilities sum or integrate to one.
- Probability zero is not impossibility, once the space is continuous.

Exercises

1. Prove that $P(A \setminus B) = P(A) - P(A \cap B)$ directly from the axioms.
2. $P(A) = 0.7$ and $P(B) = 0.6$. What are the smallest and largest possible values of $P(A \cap B)$? Justify both.
3. Show that if $A \subseteq B$ then $P(B \setminus A) = P(B) - P(A)$, and explain why this fails without the inclusion.
4. Of the shifts at a plant, 55% report a mechanical issue, 35% an electrical one and 20% both. Find the probability of: at least one; neither; exactly one; a mechanical issue only.
5. A proposed law on $\{1, 2, 3, 4\}$ is $p(k) = c(5 - k)$. Find c, then $P(k \geq 3)$.
6. Is $p(k) = k^2/50$ for $k = 1, \ldots, 6$ a probability law? If not, give the constant that would make it one.
7. Thirty components each fail in a month with probability 0.002. Bound the probability that at least one fails, and say what assumption you avoided.
8. For the three-event inclusion–exclusion formula, write out the corresponding formula for four events, and say how many terms it has.
9. A coin is tossed until the first head, with $P(k) = (1/2)^k$. Find $P(k \leq 3)$ and $P(k \text{ is odd})$.

10. On $\Omega = [0,1]$ with probability given by length, find $P([0.1, 0.4])$, $P(\{0.25\})$ and $P([0,1] \setminus [0.1, 0.4])$.

11. Give an event of probability zero that is nevertheless possible, and an event of probability one that is nevertheless not certain.

12. A student writes $P(A \cup B) = P(A) + P(B)$ for two events with $P(A) = 0.6$ and $P(B) = 0.7$. Show that the answer is impossible, and name the axiom that was misused.

Answers to the odd-numbered exercises

1. $A = (A \setminus B) \cup (A \cap B)$ and the two pieces are disjoint, so by finite additivity $P(A) = P(A \setminus B) + P(A \cap B)$; rearrange.

3. If $A \subseteq B$ then $B = A \cup (B \setminus A)$ disjointly, so $P(B) = P(A) + P(B \setminus A)$. Without the inclusion, $B \setminus A$ and A still partition $A \cup B$ rather than B, and the correct statement is $P(B \setminus A) = P(B) - P(A \cap B)$.

5. $\sum_{k=1}^{4} c(5-k) = c(4+3+2+1) = 10c = 1$, so $c = 0.1$. Then $P(k \geq 3) = c(2+1) = 0.3$.

7. $P(\text{at least one}) \leq 30 \times 0.002 = 0.06$. No assumption of independence between component failures was needed, which matters because components in one machine often fail together.

9. $P(k \leq 3) = 1/2 + 1/4 + 1/8 = 7/8$. For odd k, the series has first term $1/2$ and ratio $1/4$, giving $(1/2)/(1 - 1/4) = 2/3$.

11. With the pointer on $[0,1]$: the pointer stopping exactly at 0.5 has probability zero yet is possible; the pointer stopping anywhere other than 0.5 has probability one yet is not certain.

Before next week

Attempt exercises 1 to 7. Week 3 builds the first substantial tool on these foundations: conditional probability, the total probability theorem and Bayes' rule, which is where probability starts to say things that surprise people.